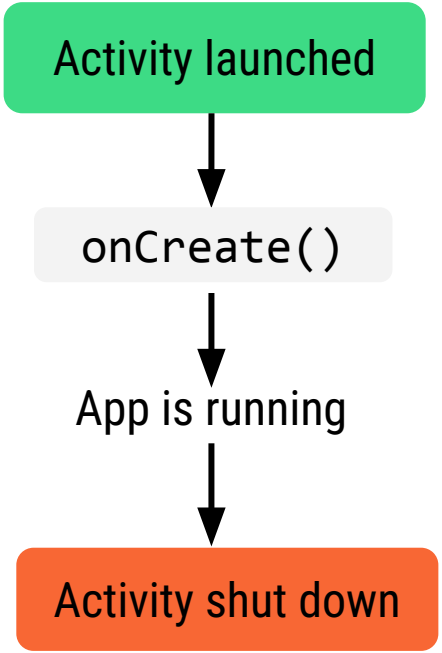
Activity and Fragment Lifecycles

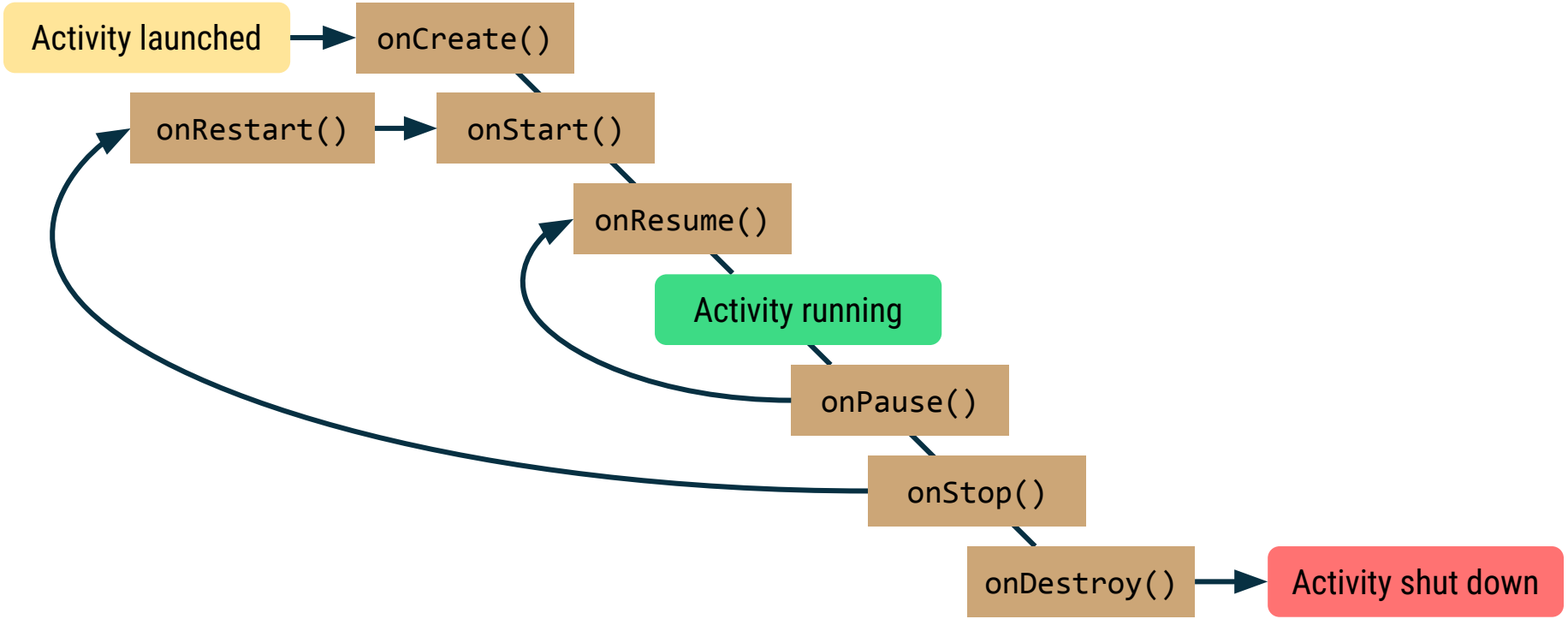
Why it matters

- Preserve user data and state if:
  - User temporarily leaves app and then returns
  - User is interrupted (for example, a phone call)
  - User rotates device
- Avoid memory leaks and app crashes.

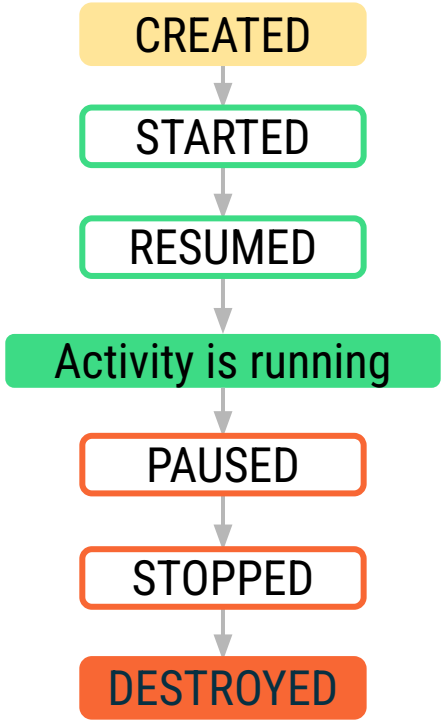
Simplified activity lifecycle



Activity lifecycle



Activity states



onCreate()

- Activity is created and other initialization work occurs
- You must implement this callback
- Inflate activity UI and perform other app startup logic

onStart()

- Activity becomes visible to the user
- Called after activity:
  - `onCreate()`

    or
  - `onRestart()` if activity was previously stopped

onResume()

- Activity gains input focus:
  - User can interact with the activity
- Activity stays in resumed state until system triggers activity to be paused

onPause()

- Activity has lost focus (not in foreground)
- Activity is still visible, but user is not actively interacting with it
- Counterpart to `onResume()`

onStop()

- Activity is no longer visible to the user
- Release resources that aren’t needed anymore
- Save any persistent state that the user is in the process of editing so they don’t lose their work

onDestroy()

- Activity is about to be destroyed, which can be caused by:
  - Activity has finished or been dismissed by the user
  - Configuration change
- Perform any final cleanup of resources.
- Don’t rely on this method to save user data (do that earlier)

Summary of activity states

State	Callbacks	Description
Created	`onCreate()`	Activity is being initialized.
Started	`onStart()`	Activity is visible to the user.
Resumed	`onResume()`	Activity has input focus.
Paused	`onPause()`	Activity does not have input focus.
Stopped	`onStop()`	Activity is no longer visible.
Destroyed	`onDestroy()`	Activity is destroyed.

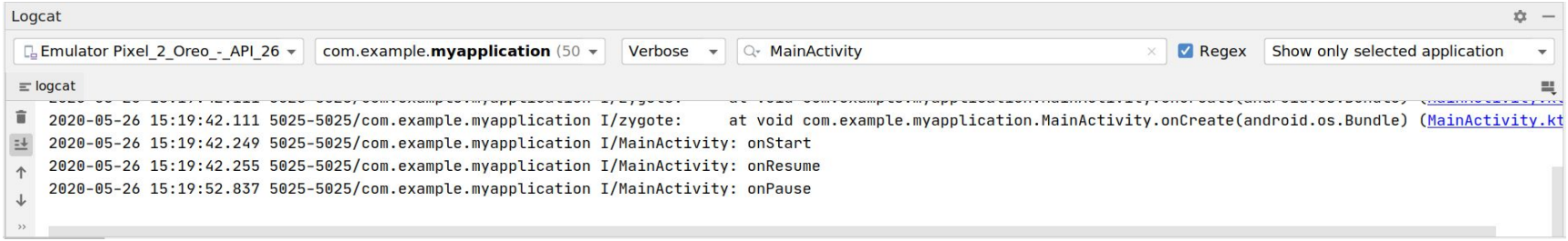
Save state

User expects UI state to stay the same after a config change or if the app is terminated when in the background.

- Activity is destroyed and restarted, or app is terminated and activity is started.
- Store user data needed to reconstruct app and activity Lifecycle changes:
  - Use `Bundle` provided by `onSaveInstanceState()`.
  - `onCreate()` receives the `Bundle` as an argument when activity is created again.

Logging in Android

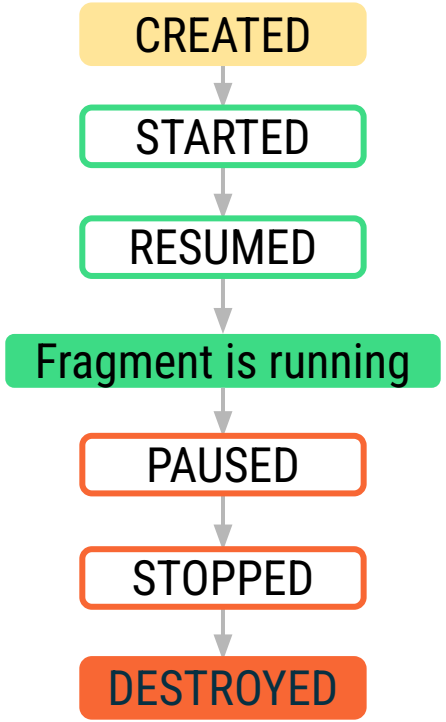
- Monitor the flow of events or state of your app.
- Use the built-in `Log` class or third-party library.
- Example `Log` method call: `Log.d(TAG, "Message")`



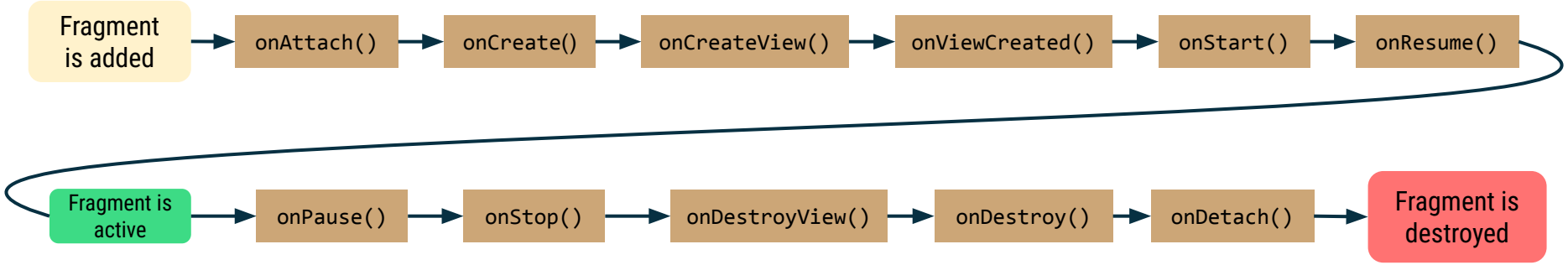
Write logs

Priority level	Log method
Verbose	`Log.v(String, String)`
Debug	`Log.d(String, String)`
Info	`Log.i(String, String)`
Warning	`Log.w(String, String)`
Error	`Log.e(String, String)`

Fragment states



Fragment lifecycle diagram



onAttach()

- Called when a fragment is attached to a context
- Immediately precedes `onCreate()`

onCreateView()

- Called to create the view hierarchy associated with the fragment
- Inflate the fragment layout here and return the root view

onViewCreated()

- Called when view hierarchy has already been created
- Perform any remaining initialization here (for example, restore state from `Bundle`)

onDestroyView() and onDetach()

- `onDestroyView()` is called when view hierarchy of fragment is removed.
- `onDetach()` is called when fragment is no longer attached to the host.

Summary of fragment states

State	Callbacks	Description
Initialized	`onAttach()`	Fragment is attached to host.
Created	`onCreate()`, `onCreateView()`, `onViewCreated()`	Fragment is created and layout is being initialized.
Started	`onStart()`	Fragment is started and visible.
Resumed	`onResume()`	Fragment has input focus.
Paused	`onPause()`	Fragment no longer has input focus.
Stopped	`onStop()`	Fragment is not visible.
Destroyed	`onDestroyView()`, `onDestroy()`, `onDetach()`	Fragment is removed from host.

Save fragment state across config changes

Preserve UI state in fragments by storing state in `Bundle`:

- `onSaveInstanceState(outState: Bundle)`

Retrieve that data by receiving the Bundle in these fragment callbacks:

- `onCreate()`
- `onCreateView()`
- `onViewCreated()`

Lifecycle-aware components

Adjust their behavior based on activity or fragment lifecycle

- Use the `androidx.lifecycle` library
- `Lifecycle` tracks the lifecycle state of an activity or fragment
  - Holds current lifecycle state
  - Dispatches lifecycle events (when there are state changes)

LifecycleOwner

- Interface that says this class has a lifecycle
- Implementers must implement `getLifecycle()` method

  Examples: `Fragment` and `AppCompatActivity` are implementations of `LifecycleOwner`

LifecycleObserver

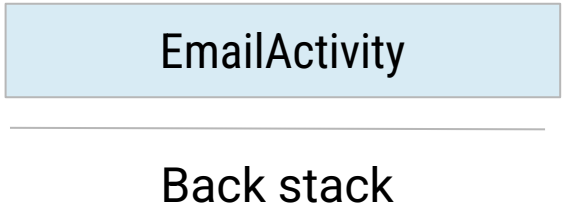
Implement `LifecycleObserver` interface:

```
class MyObserver : LifecycleObserver {
    @OnLifecycleEvent(Lifecycle.Event.ON_RESUME)
    fun connectListener() {
        ...
    }
```

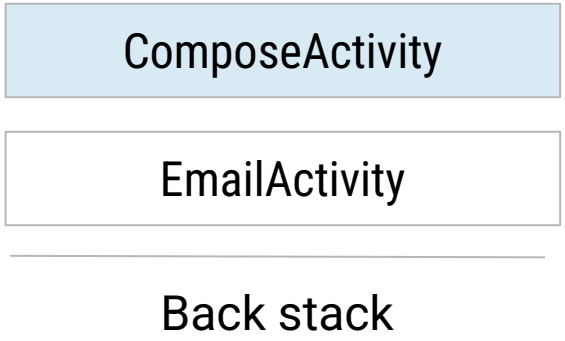
Add the observer to the lifecycle:

```
myLifecycleOwner.getLifecycle().addObserver(MyObserver())
```

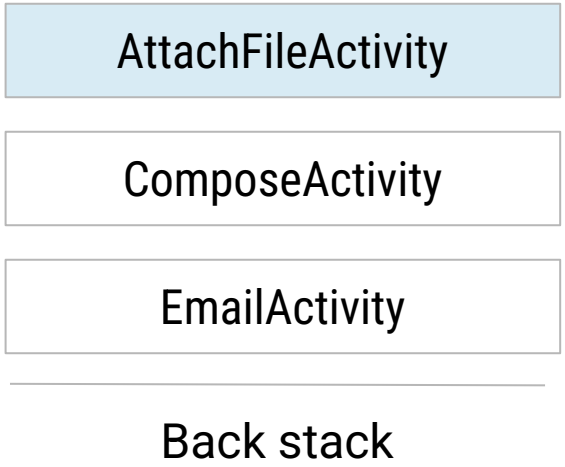
Back stack of activities



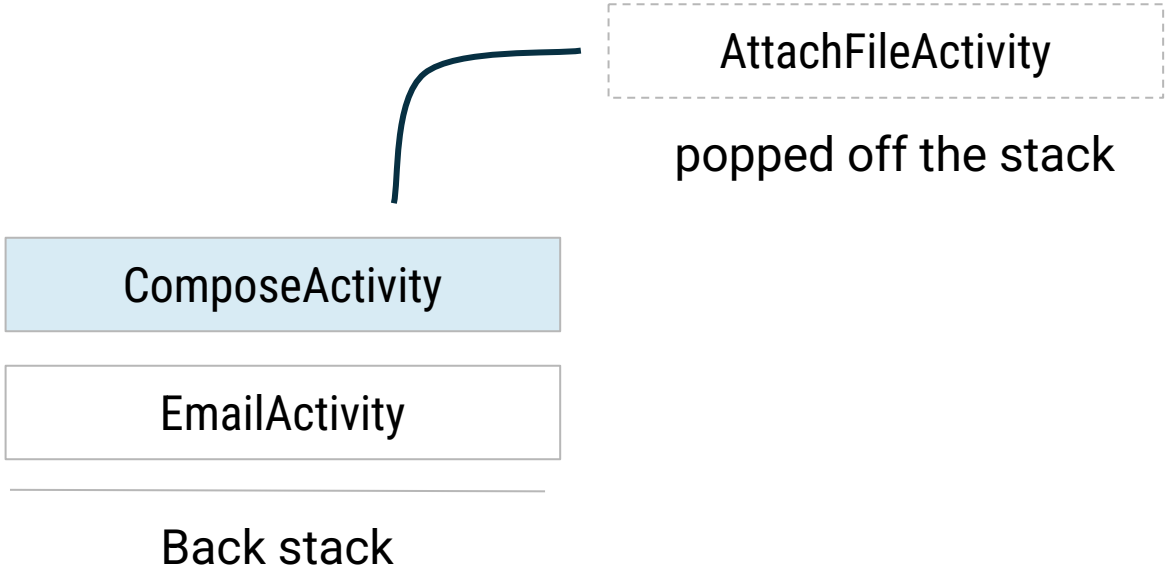
Add to the back stack



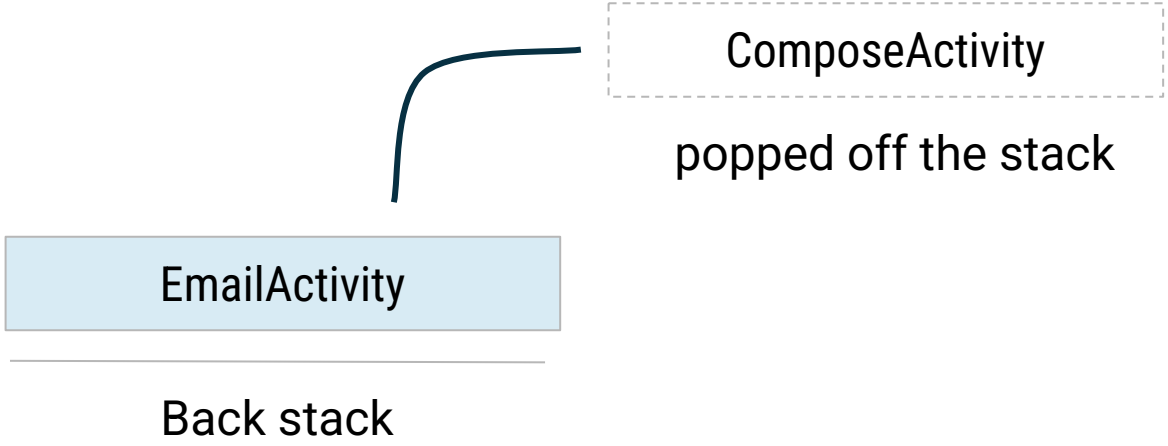
Add to the back stack again



Tap Back button



Tap Back button again



First destination in the back stack

FirstFragment

Back stack

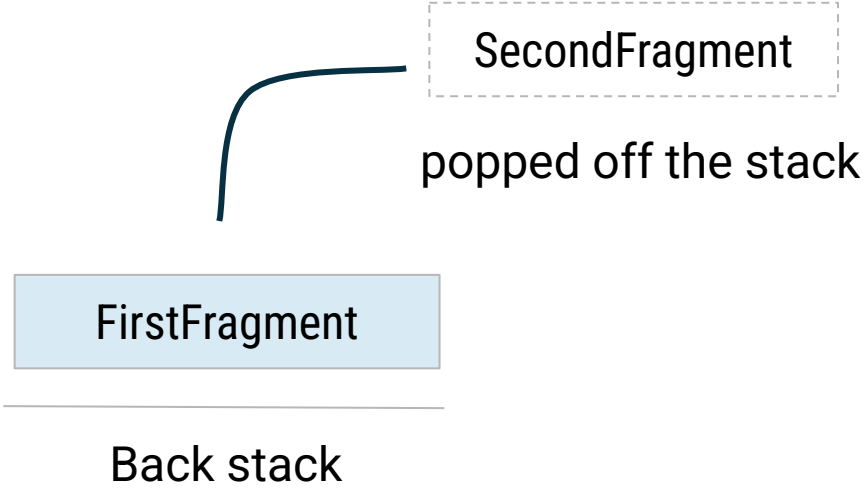
Add a destination to the back stack

SecondFragment

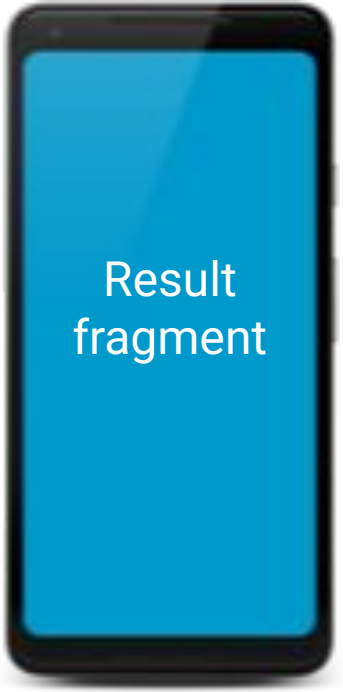
FirstFragment

Back stack

Tap Back button



Another back stack example



ResultFragment

Question3Fragment

Question2Fragment

Question1Fragment

WelcomeFragment

Back stack

Modify Back button behavior

pop additional destinations
off the back stack

ResultFragment

Question3Fragment

Question2Fragment

Question1Fragment

WelcomeFragment

Back stack